

Silverline House — 25-Storey Residential HRB

Structural General Arrangement Drawings

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1 Introduction

This document accompanies the general arrangement drawing set for the structural frame of Silverline House. The drawings are referenced SLH-STR-001 to SLH-STR-048 Rev A and are listed in the Drawing Register at Appendix A. The scope of this document covers the primary structural system, including the reinforced concrete frame, post-tensioned flat slab floors, core walls, and the ground-bearing and piled foundation system. The document does not cover secondary structures such as facade support brackets (addressed separately in the Facade Interface and Support Design document), internal partition framing, or fit-out works. All dimensions and coordinates are in millimetres unless otherwise stated. Grid references correspond to the structural grid shown on drawing SLH-STR-001 Rev A.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

2 Structural Grid and Geometry

The structural grid is on a 7.5 m × 7.5 m module aligned with the architectural grid. The core is located at the centre of the building with dimensions of 12 m × 8 m. The transfer slab at Level 1 spans 7.5 m in both directions and is 600 mm deep. Slab thicknesses above Level 1 are 250 mm flat slab throughout. Design loads have been derived in accordance with BS EN 1991-1-1 and BS EN 1991-1-3 for imposed, snow and wind loading respectively, with the UK National Annex applied throughout. The imposed load for residential floors is 1.5 kN/m² with an additional 1.0 kN/m² for partitions. Plant rooms at roof level are designed for 5.0 kN/m². Wind loading has been calculated for a basic wind speed of 23 m/s (10-minute mean at 10 m height in open country) adjusted for the site location and topography in accordance with BS EN 1991-1-4 and the UK National Annex. The net lateral wind force on the building is 3,200 kN at the 50-year return period, resisted by the reinforced concrete core.

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3 Foundation Layout

A site investigation was carried out in accordance with BS EN 1997-2 and BS 5930. The investigation comprised twelve rotary core boreholes to a maximum depth of 30 m below existing ground level, together with in-situ standard penetration tests at 1.5 m intervals. Laboratory testing was carried out on selected samples to determine soil classification, shear strength parameters, consolidation characteristics, and chemical contamination. The ground conditions comprise approximately 3 m of made ground overlying Terrace Gravels (3–9 m depth), London Clay (9–22 m), and Lambeth Group deposits below 22 m. Groundwater was encountered at 2.5 m depth in the gravels. The design groundwater level for structural purposes is taken as 0.5 m below existing ground level. The foundation system consists of 600 mm diameter continuous flight auger (CFA) piles installed to a minimum depth of 18 m below existing ground level, penetrating at least 3 m into the Lambeth Group. Pile design has been carried out in accordance with BS EN 1997-1 using a combination of ground model analysis and empirical correlations from the site investigation data. The working load per pile is 2,800 kN in compression and 800 kN in tension. Settlement analysis indicates maximum pile head settlements of 8 mm under working loads, with differential settlement between adjacent piles not exceeding 5 mm. Pile load testing in accordance with BS EN ISO 22477-1 is required during construction, with at least two maintained load tests to 150% of the working load prior to commencing pile cap construction.

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